

lab program 10 Spark Streaming Program (Scala).docx

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Summary

This program demonstrates how to process live data using **Spark Streaming** to perform automated **natural language processing** on incoming text. By establishing a connection to a local network port, the system captures raw strings and subjects them to a **multi-stage cleaning pipeline** that removes noise like punctuation and common stop words. The script further refines the data through **basic lemmatization**, which strips suffixes from words to reveal their underlying roots in real-time. Ultimately, the code serves as a functional template for transforming a chaotic **stream of words** into a standardized, readable format that is immediately displayed on the console.

Keywords: Spark Streaming, Scala programming, Text data cleaning, Socket data source, Real-time processing

Content

```
Spark Streaming Program (Scala)import org.apache.spark.sql.SparkSessionimport org.apache.spark.sql.functions._// Create Spark Sessionval spark = SparkSession.builder .appName("TextCleaningStream") .master("local[*]") .getOrCreate()import spark.implicits._// Read streaming data from socketval lines = spark.readStream .format("socket") .option("host", "localhost") .option("port", 9999) .load()// Convert to Dataset[String]val words = lines.as[String]// Define simple stop wordsval stopWords = Seq("is", "the", "a", "an", "and", "to")// Cleaning functionval cleaned = words.map { line => // Lowercase val lower = line.toLowerCase // Remove extra spaces val noSpace = lower.trim.replaceAll("\\s+", " ") // Remove punctuation val noPunct = noSpace.replaceAll("[^a-zA-Z ]", "") // Remove stop words val filtered = noPunct.split(" ") .filter(word => !stopWords.contains(word)) // Simple lemmatization (basic example) val lemmatized = filtered.map { case w if w.endsWith("ing") => w.dropRight(3) case w if
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w.endsWith("ed") => w.dropRight(2) case w => w } lemmatized.mkString(" ")}
```

Output to consoleval query =

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cleaned.writeStream .outputMode("append") .format("console") .start()query.awaitTermination()
```

How to RunStart Spark applicationOpen terminal and run:nc -lk 9999Type text →

cleaned output appears in Spark console